

Structural Persistence: Minimal Kernel and Balance Principle

Feasible-region shrinkage, log-ratios, exponential kernels, and recovery-aware persistence

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要旨

A structure can become unmaintainable even when resources remain. This paper separates that failure into a resource-side effective maintenance surplus M and shrinkage of the state region compatible with the target structural condition, read as L/B .

This paper is a reader-facing synthesis of two companion theoretical papers: The Minimal Form of Structural Persistence and The Balance Principle of Structural Persistence. The precise axioms, proofs, and limitations remain those of Paper 1 / Paper 2 and the corresponding technical supplements.

In the closed shrinkage case, a pre-fixed feasible region V and ruler m yield the exponential kernel e^{-L} by reading survival ratios on a logarithmic scale. With the resource-side scalar M made explicit, the structural persistence potential is written $S = Me^{-L}$.

For open systems with repair, recovery, reconfiguration, or external support, each time step has structural consumption d_t and recovery r_t . Their difference $b_t = d_t - r_t$ accumulates to B_n , giving the recovery-aware form $S_n = M_n e^{-B_n}$. The novelty is not the exponential notation itself. It is the pre-fixed coordinate that separates resource-side surplus from feasible-region shrinkage, making it possible to describe how a structure becomes unmaintainable even when resources remain.

1 Role of this paper

This paper is the integrated reading path for the main theoretical spine.

Document	Role
Paper 0	Whole-program map: observability layers, evidence, Lean, supplements
This Core Paper	Reader-facing synthesis of Paper 1 and Paper 2
Paper 1	Strict spine for measuring shrinkage, log-ratio uniqueness, telescoping product, and the minimal kernel
Paper 2	Strict spine for two-step updates, recovery, net consumption, and the balance principle

This paper is therefore not a replacement for the companion papers. It is the shortest public path through them.

Paper 1 is not merely “obvious.” It fixes the object that is being measured: the set of states in which a structure can still be maintained. It then shows that, if shrinkage is measured by a scale that composes additively over chained survival ratios, the log-ratio scale is unique. It also explains why post-hoc choices of the feasible set or ruler m would make the formalism empty.

2 The question

This theory does not begin by projecting abstract mathematics onto reality. In businesses, organizations, institutions, development practice, learning systems, and reasoning systems, a structure can become unmaintainable even when resources and slack still remain. The starting question of this paper is whether that phenomenon can be formalized not as simple resource shortage, but as shrinkage of the state region compatible with the target structural condition.

This paper fixes a function or structural condition to be maintained on a target system X . It then considers the region of states in which X remains compatible with that structural condition.

Intuitively, an organization may still have people and budget, yet lose the coherent decision states or executable procedures that let it continue as that organization. Software may still run and have compute resources, yet lose the states in which changes remain coherent or recovery paths remain available. What is

lost is not existence in general, but the state region in which the system remains compatible with the target structural condition.

More concretely, the question is why a system can become unable to maintain its structure even when resources remain. Is there a descriptive coordinate that can be reused across domains?

A structure can become unmaintainable even when resources remain. If the state region narrows and no state satisfies the maintenance conditions, the system can no longer remain compatible with that structural condition.

The boundary at which a system can no longer remain compatible with the target structural condition, the structural maintenance boundary, may appear differently across domains: collapse, functional failure, operational halt, phase transition, or structural reorganization. These are not treated as separate universal phenomena. Under a pre-fixed structural-maintenance map, they may be read as different manifestations of the same boundary. What can be separated is the route to that boundary. One route is shrinkage of the feasible region, read on the V, L, B side. Another route is exhaustion of effective maintenance surplus M , which may appear as failure or halt even when some feasible structural region remains. The minimal kernel formalizes the former and keeps M explicit as the resource-side effective maintenance surplus.

Thus the question is not simply how much resource remains, but how the region of states compatible with the target structural condition changes under constraints, contradictions, damage, updates, disturbances, repair, or reconfiguration.

This question splits into two parts.

1. Without recovery, how should we measure shrinkage of the state region in which the system remains compatible with that structural condition?
2. With recovery, how should structural consumption and recovery be balanced on the same scale?

Paper 1 answers the first question. Paper 2 answers the second. This paper connects them.

3 What must be fixed before measuring structure

Let X be a system and let V denote the states compatible with maintaining the target structural condition. Write the initial feasible region as $V^{(0)}$. Under accumulating constraints,

$$V^{(0)} \supseteq V^{(1)} \supseteq \dots \supseteq V^{(n)}.$$

A ruler m is fixed to compare the sizes of these feasible regions. Mathematically, m is a finite measure; reader-facing, it is the convention used to read the size of a state set. It may be counting measure in a finite system, probability mass under a pre-fixed distribution, volume or weighted volume in a continuous state space, or another pre-specified finite measure. If a coordinate transform or feature extraction changes the meaning of m , that map must also be fixed in advance. The detailed discipline for choosing this ruler and freezing the corresponding map is only summarized here; it is specified in Section 13 and the v3 operations files.

The nontrivial step is not the act of taking a logarithm. It is fixing, before seeing the outcome, which state set counts as structurally feasible and which ruler m is used to read its mass. The same observed system can yield different L values if one post-hoc switches between a consistency set, a success set, or a function-maintenance set. If V or m can be chosen after seeing the data, almost any monotone sequence can be represented and the theory loses empirical content. This pre-fixing of V , m , the stage sequence, and the time horizon is therefore not a bookkeeping prelude; it is what turns the formalism into an empirical claim.

All log-ratios below are read on positive finite masses. Hitting zero is treated separately as a collapse boundary, not as an ordinary finite log-ratio update.

As the simplest specification-fixed example, consider a finite CSP. Let X be the set of all candidate assignments, let $V^{(i)}$ be the candidate set satisfying the first i constraints, and let $m(V) = |V|$ be the number of candidates. Then d_i reads how much constraint i removed from the maintainable candidate set. If relaxation or repair expands the candidate set again, that expansion is read, in Paper 2, as r_t on the same log-ratio scale.

4 How to measure shrinkage

Let $r \in (0, 1]$ be a survival ratio. A structural consumption scale $f(r)$ should depend on the already-realized ratio, not on the absolute mass. The composition requirement is not a probabilistic independence assumption about how constraints are generated.

It is a measurement requirement. If a shrinkage path factors as

$$\begin{aligned} \frac{m(V^{(2)})}{m(V^{(0)})} &= \\ \frac{m(V^{(1)})}{m(V^{(0)})} & \\ \frac{m(V^{(2)})}{m(V^{(1)})}, & \end{aligned}$$

then the measured consumption should add:

$$f(r_1 r_2) = f(r_1) + f(r_2).$$

Overlap, dependence, and redundancy among constraints are already absorbed into the actual feasible sets $V^{(i)}$ and the ratios $m(V^{(i)})/m(V^{(i-1)})$. The additivity axiom concerns the scale used to measure those ratios.

Together with normalization and continuity, this functional equation forces

$$f(r) = -k \log r.$$

Taking $k = 1$ fixes the unit. The stage consumption is

$$\begin{aligned} d_i &= \\ -\log \frac{m(V^{(i)})}{m(V^{(i-1)})}. & \end{aligned}$$

5 Accumulating survival ratios yields the exponential kernel

Define cumulative structural consumption by

$$L_n = \sum_{i=1}^n d_i.$$

Then the intermediate ratios cancel in a telescoping product:

$$m(V^{(n)}) = m(V^{(0)})e^{-L_n}.$$

This is not the empirical claim that every real system decays exponentially. It is a representation theorem: once a pre-fixed feasible region is measured through log-ratios, the exponential kernel follows algebraically.

The law-side kernel is therefore e^{-L} . Introducing the resource-side term M , the structural persistence potential is

$$S = Me^{-L}.$$

Here S is the structural persistence potential for the target structural condition. In general, it is not itself a probability or a physical observable; it is a comparison quantity defined under pre-fixed choices of M, V, m , and the boundary convention. Reaching $S = 0$, or a pre-fixed threshold $S \leq S_c$, means reaching the structural maintenance boundary for that condition. For finite L or B , e^{-L} and e^{-B} remain positive, so $S = 0$ is read mainly as $M = 0$, a zero-mass boundary, or a limiting boundary. This is not the empirical claim that physical collapse must occur. Observationally, the boundary may appear as collapse, functional failure, halt, phase transition, or structural reorganization, depending on the domain.

The term M is not derived from the exponential kernel. It represents the resource-side effective maintenance surplus available for the target structural condition, and is treated here as a pre-fixed resource-side readout. This does not assume that M and L/B vary independently in the real dynamics. A shortage of effective maintenance surplus can act as a constraint that affects future feasible regions V , recovery terms r_t , cumulative net consumption B , or observed boundaries. The structural maintenance boundary can be reached through the feasible-region route represented by L , the resource-side route represented by $M = 0$, or both. The new law-side coordinate is L , the shrinkage of the feasible region.

6 What changes when recovery is included

Open systems may repair, restore, buffer, reroute, roll back, or reconfigure. A time step can therefore be read as a two-step update:

$$V_t^- := K_t(V^{(t)}),$$

$$V^{(t+1)} := R_t(V_t^-).$$

Here K_t is the shrinkage action and R_t is the recovery action within the fixed observation unit.

Define

$$\begin{aligned} d_t &= \\ &= -\log \frac{m(V_t^-)}{m(V^{(t)})}, \end{aligned}$$

$$\begin{aligned} r_t &= \\ &= \log \frac{m(V^{(t+1)})}{m(V_t^-)}. \end{aligned}$$

Then

$$\begin{aligned} b_t &= d_t - r_t \\ &= \\ &= -\log \frac{m(V^{(t+1)})}{m(V^{(t)})}. \end{aligned}$$

The intermediate set cancels in the log-ratio accounting. This cancellation is an accounting identity for a given start point, intermediate point, and endpoint. It is not a claim that the physical order of consumption and recovery never matters. If changing the order changes the endpoint $V^{(t+1)}$, then b_t changes as well. What is order-invariant here is the log-ratio reading, not the underlying dynamics.

If reconfiguration changes the target structural condition itself, the old and new structures may not be directly comparable with the same ruler. In that case, a

common map or common embedding must be fixed in advance before the two structures are read in the same coordinate.

7 The balance kernel

Define

$$B_n = \sum_{t < n} b_t.$$

The same telescoping argument gives

$$m(V^{(n)}) = m(V^{(0)})e^{-B_n}.$$

With the resource-side term,

$$S_n = M_n e^{-B_n}.$$

If recovery exceeds cumulative consumption, then $B_n < 0$ and $e^{-B_n} > 1$. Under the pre-fixed ruler m , this means that the feasible region has expanded beyond the initial reference level.

If there is no recovery, then $r_t = 0$, $b_t = d_t$, and $B_n = L_n$. Thus the balance kernel does not replace the minimal kernel. It extends it.

The central point is simple: in recovery-aware systems, persistence is governed not by consumption alone, but by net consumption.

8 Accounting is not equilibrium

The word “balance” means accounting, not equilibrium. The signs of b_t give local accounting regimes on a pre-fixed observation unit.

Condition	Reading
(b _t >0)	consumption-dominant
(b _t =0)	locally maintained
(b _t <0)	recovery-dominant

These regimes are grain-relative. If two adjacent intervals are merged,

$$\begin{aligned}
& b_{t:t+2} \\
& = \\
& -\log \frac{m(V^{(t+2)})}{m(V^{(t)})} \\
& = \\
& b_t + b_{t+1}.
\end{aligned}$$

The cumulative quantity B_n is additively coherent under time aggregation, but regime labels and hitting boundaries depend on the pre-fixed observation unit. A path that is “consumption followed by recovery” at a fine grain may appear locally maintained at a coarser grain.

9 Observability layers

The same kernel is read through three observability layers.

Layer	Reading
Specification-fixed structural layer	Structure, ruler (m), and boundary are directly fixed by the specification
Conditional structural-embedding layer	Existing drift, difference, or hitting-bound theories are mapped conditionally into the structural variables
Structurally inferred layer	The structure is not directly counted; observation and inference indicators are frozen and tested

These are not different theories. They are different observational interfaces to the same kernel.

The law-side claim belongs to the specification-fixed structural layer. The structurally inferred layer is an engineering layer: it uses observation and inference indicators to approximate the law-side coordinates L/B . The resource-side term M is kept outside as a scalar condition. The quality of the L/B approximation is judged only after freezing the mapping and testing whether the structural-

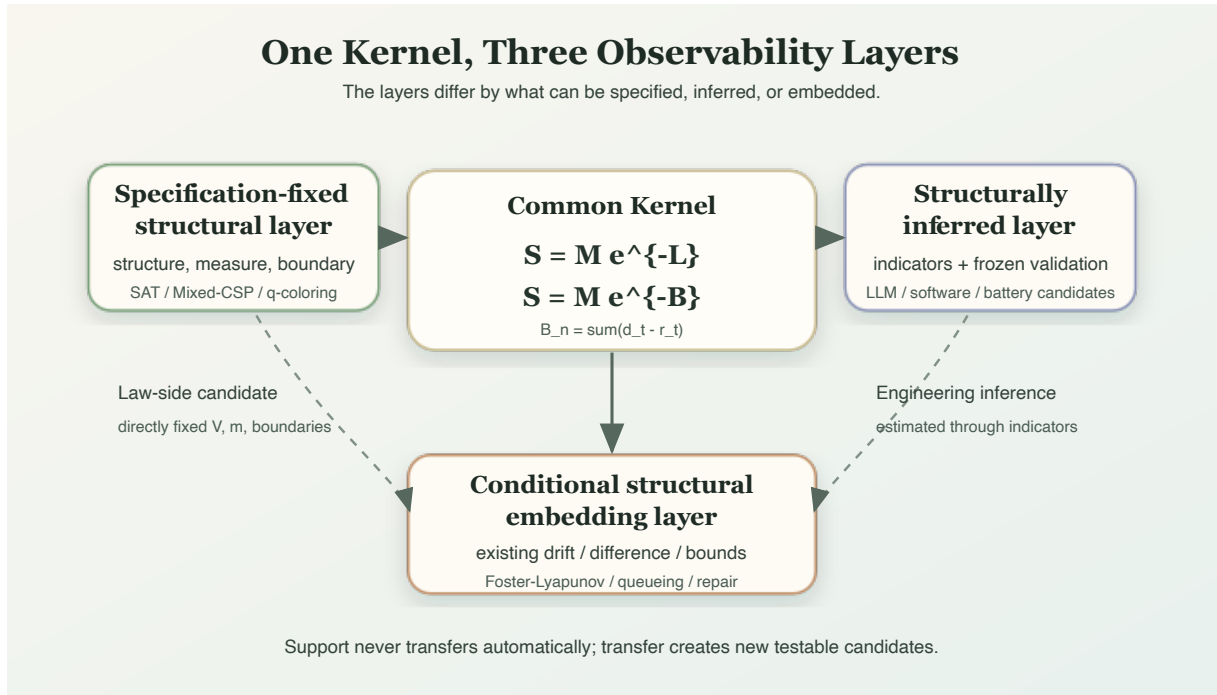


Figure 1: Figure 1. One structural persistence kernel read through three observability layers.

persistence indicators add out-of-sample predictive value beyond the domain baseline.

The fact that L and B are dimensionless log-ratios, readable in natural-log units, does not by itself justify quantitative comparison across domains. Cross-domain comparison becomes meaningful only under pre-fixed maps specifying V , the ruler m , the observation unit, and the boundary in each domain. Under those conditions, L/B provide a candidate common coordinate for shrinkage and recovery.

As the approximation improves, the structural persistence coordinates move from explanatory vocabulary toward predictive instrumentation. This is an empirical achievement, not something granted by the notation.

10 What the Lean formalization covers

The Lean formalization does not prove the entire theory. Its role is to check the algebraic skeleton used by the main theory and the common interface used by registered limited classes.

The key point is that three different registered limited classes—Bernoulli-CSP,

Foster-Lyapunov / queueing, and Repair-Maintenance—are checked in Lean as satisfying the same unifying interface. This is not a single universal inequality for all systems. It is a machine-checked fact that multiple law-side candidate classes can be organized under the same structural-persistence vocabulary.

The core Lean-covered parts are, roughly:

1. the minimal algebra behind log-ratios, telescoping products, and the exponential kernel;
2. the balance algebra using d_t , r_t , $b_t = d_t - r_t$, and B_n ;
3. the fact that registered limited classes such as Bernoulli-CSP, Foster-Lyapunov / queueing, and Repair-Maintenance satisfy a common unifying interface.

Lean does not by itself prove that every domain has a natural feasible region V , ruler m , or observation unit. It also does not prove that structurally inferred indicators approximate the true L/B , which real resource should be read as M , that an empirical package obtains outside-rerun support, or that a single universal law holds for all systems.

Thus Lean supports the law-side skeleton. Empirical support and structurally inferred validity are still judged by frozen validation, holdout tests, outside reruns, and the support / no-support / silence discipline.

11 Evidence status

The current evidence should be read by strength.

- **Hard evidence:** the hard footing lies in the specification-fixed structural layer. Mixed-CSP and q-coloring each have frozen empirical packages with three outside rerunners reproducing decision-relevant outputs. Mixed-CSP has 12,000 primary rows per returned run, 0 checked core mismatches, and reproduced support flags. q-coloring has 4,000 primary rows per returned run, 0 checked core mismatches, $TIMEOUT = 0$, $MALFORMED = 0$, and the qualitative support decision reproduced. This is not a proof of the whole theory, but it is the strongest current package-scoped outside-rerun support for the law-side candidate.
- **Auxiliary and exploratory evidence:** in the structurally inferred layer, LLM reasoning, continual learning, and software diagnostics provide evidence

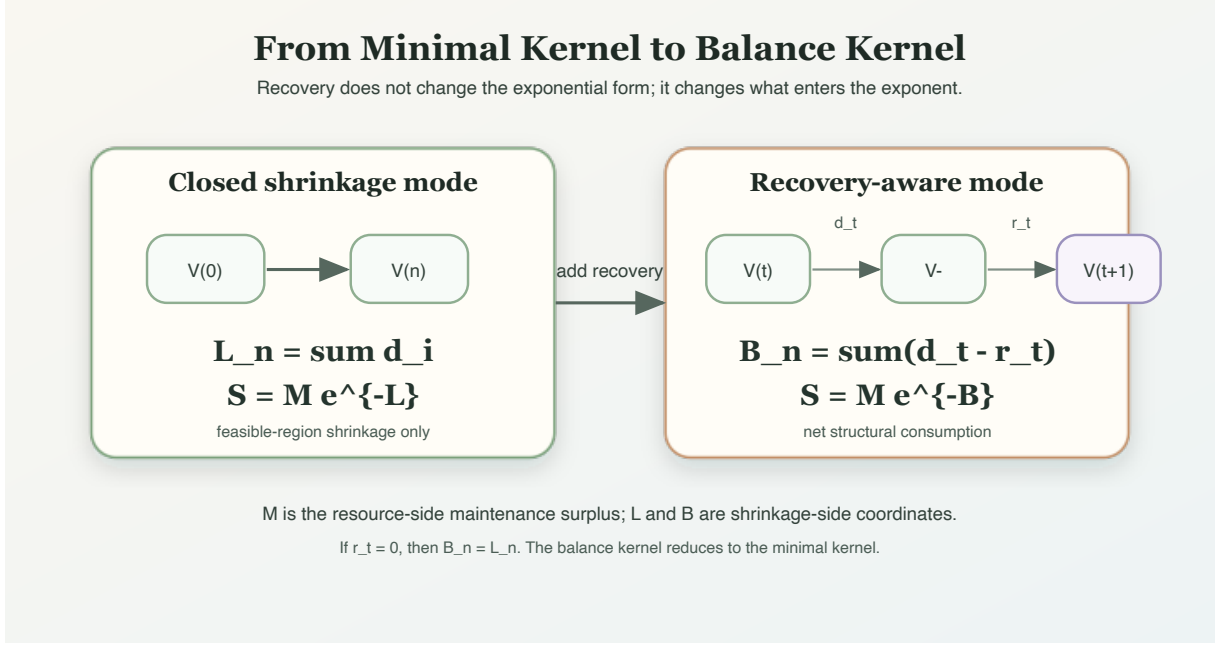


Figure 2: Figure 2. The minimal kernel lifts to the balance kernel by replacing cumulative consumption L with cumulative net consumption B .

through frozen observation and inference indicators. The question there is whether the indicators add information beyond the domain baseline, not whether the true $V, m, L/B$ have been directly counted.

- **Theoretical anchors:** in the conditional structural-embedding layer, Foster-Lyapunov / queueing, Repair-Maintenance, reliability, and decay systems act as anchors. They map existing drift, difference, and boundary arguments into the structural-persistence vocabulary; they are not new empirical wins.
- **Evidence ledger:** Backblaze, C-MAPSS, Scania, Oxford battery, and the M-flow network testbed belong to the ledger of support, weakening, and no-support. They are recorded to preserve where the framework works, weakens, or fails.

12 What this makes possible

The value of this core paper is not merely that it places two formulas side by side. The coordinate makes it possible to separate routes to unmaintainability into a resource-side route and a feasible-region shrinkage route.

First, the theory separates the support side from the shrinkage side. The familiar resource-side term is kept as effective maintenance surplus M , while the loss

of states compatible with the target structural condition is read as L , or as B when recovery is present. This makes it possible to read collapse, functional failure, halt, phase transition, and structural reorganization as possible manifestations of a structural maintenance boundary, while still distinguishing the route by which that boundary is reached.

Second, L and B provide candidate common coordinates for structural consumption and recovery across domains. They are dimensionless log-ratios, readable in natural-log units when $k = 1$. This unit convention alone does not justify cross-domain comparison. Comparison becomes meaningful only under pre-fixed maps specifying V , the ruler m , the observation unit, and the boundary in each domain.

Third, the framework separates where law-side claims are made from where observation or inference indicators are tested. The specification-fixed structural layer can support law-side claims. The structurally inferred layer tests whether frozen indicators add predictive information beyond the domain baseline. The conditional structural-embedding layer connects existing theories to the same vocabulary. This separation keeps support, no-support, and silence distinct.

Fourth, the empirical question is not whether structural-persistence indicators are always the strongest standalone model. Many domains already have strong domain baselines. The central test is whether

domain baseline + structural-persistence indicators

improves over the domain baseline alone under frozen out-of-sample conditions.

Fifth, the coordinate gives a language for design transfer. Localization of consumption, preservation of recovery paths, preservation of safety margin, and preservation of alternative paths can become candidate interventions in another domain. What transfers is not support itself. Support in the new domain must be established by freezing the map, indicators, baseline, metric, split, and decision rule, and then testing on held-out data, future data, a fresh archive, or an outside rerun.

13 Minimal operational discipline

This paper does not reproduce the full procedure for mapping structural persistence into a concrete domain. The detailed procedure is kept in the v3 operations files: the standard mapping procedure, operational discipline, support levels, and failure ledger. Here we state only the minimal discipline needed to read this core paper.

The theory separates mapping discovery from validation. A structural condition, ruler, observation or inference indicator, or recovery indicator found during exploration is not support by itself. Support is assigned only after the structural condition, mapping, indicators, baseline, metric, data split, and decision rule have been frozen, and only if the frozen rule succeeds on unused data, a future surface, a fresh archive, or an outside rerun.

If coarse-graining, reconfiguration, or action order can change the readout of endpoints or boundaries, the observation unit, common map, embedding, and decision boundary are also part of what must be frozen.

In the specification-fixed structural layer, one first fixes the system X , the function or structural condition to be maintained, the feasible region V , the ruler m , the observation unit, the time horizon, and the structural-maintenance boundary. In the structurally inferred layer, one does not claim to directly count the true $V, m, L/B$; one tests whether frozen observation or inference indicators add information beyond the domain baseline. In the conditional structural-embedding layer, one maps existing differences, drifts, stopping boundaries, and stability conditions into the L/B vocabulary.

A frozen mapping that fails is recorded as no-support. It is not promoted to support by retuning the indicator or boundary on the same data. Failures are kept as a ledger of where a mapping failed and where the theory should remain silent.

14 Conclusion

The law-side minimal kernel is e^{-L} . With resource-side effective maintenance surplus made explicit, the structural persistence potential is

$$S = Me^{-L}.$$

The recovery-aware balance kernel replaces L by cumulative net structural consumption B :

$$S = Me^{-B}.$$

The contribution is not the statement that resources matter. The contribution is the separation between the resource-side term M and the structural shrinkage coordinates L and B . This coordinate lets one read routes to unmaintainability as either resource-side exhaustion or feasible-region shrinkage. It also makes it possible to say when a structure fails because the feasible region compatible with that structure has been consumed, even if resources have not simply vanished.